

QUEBEC CITY BEAUPORT, QC, Canada

WMO: 715780

Lat: 46.8371N

Lon: 71.1973W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 99-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-4.9	-21.5	1.7	-8.7	-16.3	2.3	-4.0	29.7	22.1	26.3	19.9	11.0	240	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	85.9	70.0	82.8	68.2	79.8	66.8	73.5	81.7	71.2	79.0	69.3	76.3	11.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.8	113.6	78.1	68.6	105.3	75.5	66.6	98.1	73.6	37.1	82.1	35.1	79.1	33.4	76.3	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	23.6	20.7	DB	-14.9	91.0	5.8	3.7	-19.0	93.7	-22.4	95.9	-25.7	97.9	-29.9	100.6	(4)
(5)			WB	-16.0	75.8	5.2	1.9	-19.8	77.2	-22.8	78.3	-25.8	79.3	-29.6	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	13.9	17.4	27.1	40.2	53.3	63.0	69.2	67.5	59.7	46.6	35.5	21.6		
	DBStd	20.65	11.64	9.89	9.46	6.80	7.02	6.45	5.27	5.25	6.50	7.29	7.26	9.74		
	HDD50	4561	1120	912	711	301	39	0	0	0	3	156	437	882		
	HDD65	8357	1585	1332	1175	743	369	114	18	34	183	572	885	1347		
	CDD50	2025	0	0	0	8	142	389	596	543	296	49	2	0		
	CDD65	348	0	0	0	0	8	53	148	112	26	1	0	0		
	CDH74	2397	0	0	0	4	127	402	991	680	183	10	0	0		
	CDH80	622	0	0	0	0	32	114	273	156	45	2	0	0		
	WSAvg	9.6	10.1	10.2	10.1	10.9	11.0	9.0	7.8	7.8	8.0	9.8	9.9	10.2		
	PrecAvg	48.9	3.4	2.9	3.0	3.7	4.0	4.6	5.2	4.9	4.7	4.5	3.9	4.0		
(15) Precipitation	PrecMax	55.9	6.0	5.8	5.8	8.7	10.3	9.4	10.2	8.3	9.1	9.6	6.5	8.1		
	PrecMin	38.6	1.3	0.9	0.9	1.3	1.8	1.8	1.7	1.1	3.1	1.9	1.1	2.0		
	PrecStd	4.8	1.3	1.1	1.2	1.9	1.7	1.6	1.9	1.5	1.3	1.8	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	40.1	53.3	68.8	84.6	88.5	91.8	88.0	85.4	74.0	56.6	43.1		
		MCWB	38.6	37.3	44.6	53.5	62.8	70.7	75.8	72.3	71.3	62.9	50.8	41.1		
	2%	DB	36.5	36.3	45.9	61.0	77.3	83.2	86.0	84.3	79.3	67.0	51.6	39.1		
		MCWB	34.7	33.8	38.6	47.2	59.6	67.5	71.5	69.3	67.4	58.0	48.1	36.7		
	5%	DB	34.0	34.0	41.4	54.7	70.8	78.6	82.7	81.0	74.2	61.3	47.7	36.7		
		MCWB	32.6	31.8	36.6	44.2	57.6	64.5	69.3	67.8	64.2	55.5	44.3	34.7		
	10%	DB	30.0	31.7	38.1	50.1	65.2	74.4	79.6	77.8	70.3	56.7	44.9	34.4		
		MCWB	28.4	30.0	34.3	42.3	54.2	62.5	67.6	66.5	62.5	51.6	42.1	32.8		
	0.4%	WB	38.9	37.8	46.6	56.6	67.9	73.0	77.4	74.6	72.7	64.6	52.9	42.0		
		MCDB	40.0	39.6	50.2	66.2	76.3	85.6	86.8	84.2	81.5	72.2	55.0	42.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	34.1	39.6	48.7	63.4	70.0	74.9	72.0	68.7	59.2	48.4	37.4		
		MCDB	36.2	35.7	44.2	57.7	71.5	79.2	82.9	79.5	74.9	65.6	50.7	38.6		
	5%	WB	32.7	32.2	37.1	45.9	58.6	67.6	72.1	70.3	65.8	56.1	44.8	35.0		
		MCDB	33.9	33.7	40.6	52.6	67.4	74.3	79.2	77.4	71.1	60.6	46.7	36.3		
	10%	WB	28.5	30.0	34.8	43.5	55.7	65.1	69.9	68.6	63.5	52.3	42.5	33.0		
		MCDB	30.2	31.6	37.7	48.7	63.8	71.9	76.7	75.4	68.6	55.8	44.7	34.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	14.2	13.8	13.3	15.5	16.2	16.7	16.4	16.3	12.4	10.2	11.2		
		MCDBR	14.5	12.9	16.4	23.4	26.4	23.6	21.7	21.4	22.2	20.6	14.7	10.8		
		MCWBR	14.4	11.1	11.8	13.9	13.7	11.8	10.9	10.8	11.7	13.3	11.9	10.3		
		MCDBR	13.8	11.7	14.7	19.4	22.2	19.5	18.9	17.7	18.1	18.5	12.9	10.4		
	5% WB	MCWBR	14.0	10.6	11.4	12.9	13.4	12.0	11.0	10.5	11.6	12.8	11.3	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.287	0.329	0.356	0.371	0.407	0.418	0.399	0.374	0.342	0.311	0.295		
	taud		2.312	2.243	2.208	2.333	2.387	2.285	2.274	2.330	2.393	2.471	2.460	2.274		
	Ebn at Noon		263	279	282	285	284	272	267	268	265	258	245	233		
	Edh at Noon		23	31	38	37	36	41	41	37	32	25	21	22		
(44) All-Sky Solar Radiation	RadAvg		445	732	1114	1409	1605	1748	1711	1487	1167	682	394	300		
	RadStd		62	90	90	121	167	111	169	94	98	57	47	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page